Statistical Learning and Machine Learning

Exercise Session (Week 6)

Book Exercises

Problem 3.7, 3.10

Linear Regression

Consider a training data set of inputs $\{t_1 = -1, t_2 = 1, t_3 = 2\}$ and the corresponding targets or outputs $\{x_1 = 1, x_2 = 1.5, x_3 = 3\}$. The goal is to first fit an appropriate linear regression model to the above data using the least squares approach and then use that model to predict the value of x for a new value of t.

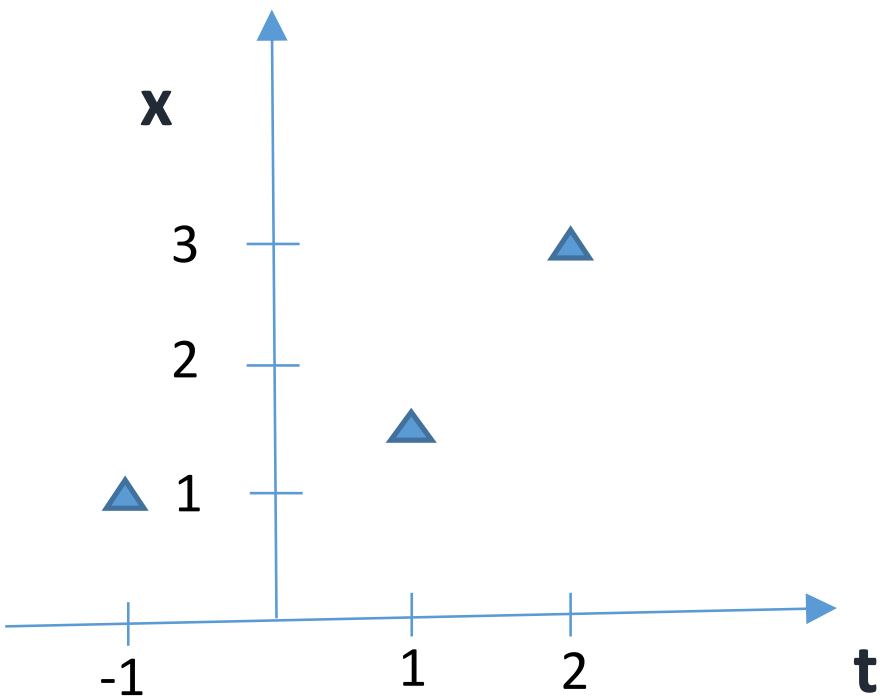

Fig. 1. The training data set consisting of $N = 3$ points (represented as blue triangles) plotted in the 2D space.

1) (Score: 25 %) Considering the following choice for the basis functions $\mathbf{\Phi}(t) = \{\Phi_0(t), \Phi_1(t)\} = \{1, t\}$ that leads to the simplest linear model for regression $x(t, w) = w_0 + w_1 t$, write down the corresponding *normal equations* for the least squares problem.
2) (Score: 25 %) Solve the normal equations to obtain the parameters or the weights (w_0 and w_1) of the linear model. Hint: The inverse of a 2×2 matrix is
$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}[r]d - b - ca\,.$$
3) (Score: 25 %) Plot the linear model function $x(t, w) = w_0 + w_1 t$, obtained in the previous step, in Fig. 1. You can assess your model by its ability to fit the training data i.e., how close it is to the training data.
4) (Score: 25 %) Using the linear model that is estimated in the previous steps, predict the value of x for $t = 0$, and for $t = 3$. Plot the corresponding points in Fig. 1 as crosses.